School: ....Campus: ....

Academic Year: .... Subject Name: .... Subject Code: ....

Semester: .... Program: .... Branch: .... Specialization: ....

Date: ....

Applied and Action Learning

(Learning by Doing and Discovery)

Name of the Experiment: Token Launch – Deploying a Token Locally

*Coding Phase: Pseudo Code / Flow Chart / Algorithm

Step 1:

Open the **Remix IDE** by visiting https://remix.ethereum.org.

Step 2:

In the **contracts** folder, click **"Create New File"** and name it **`MyToken.sol`**.

Step 3:

Inside the `.sol` file, write the **ERC-20 smart contract code** defining the token's name, symbol, total supply, and standard functions like `transfer`, `approve`, and `transferFrom`.

Step 4:

Navigate to the **Solidity Compiler** tab and **compile** the contract.

Step 5:

After successful compilation, open the **Deploy & Run Transactions** tab and select **"Injected Provider – MetaMask"** as the deployment environment.

Step 6:

Click **"Deploy"**. MetaMask will open a confirmation popup — review the transaction details and click **"Confirm"** to deploy the ERC-20 token contract to the **Sepolia Test Network**.

Step 7:

Once deployed, go to the **Deployed Contracts** section in Remix and **copy the contract address**

.

Step 8:

Open your **MetaMask wallet**, go to the **Tokens** tab, and select **“Import Tokens.”**

Step 9:

In the import section, make sure you are connected to the **Sepolia network**.

Paste the **contract address** you copied earlier, and MetaMask will automatically fetch the **token symbol** and **decimals**.

Click **“Next.”**

Step 10:

Click **“Import”** to successfully add your custom token to MetaMask.

Step 11:

Go to the **Tokens** tab again, and you will see your **imported ERC-20 token** listed.

Step 12:

To transfer tokens, click **“Send”**, enter the recipient’s **public address** (or ENS domain name), and click **“Continue.”**

Step 13:

Confirm the transaction in MetaMask.

Your **ERC-20 tokens** will be **sent successfully** once the transaction is processed on the blockchain.

*software used:

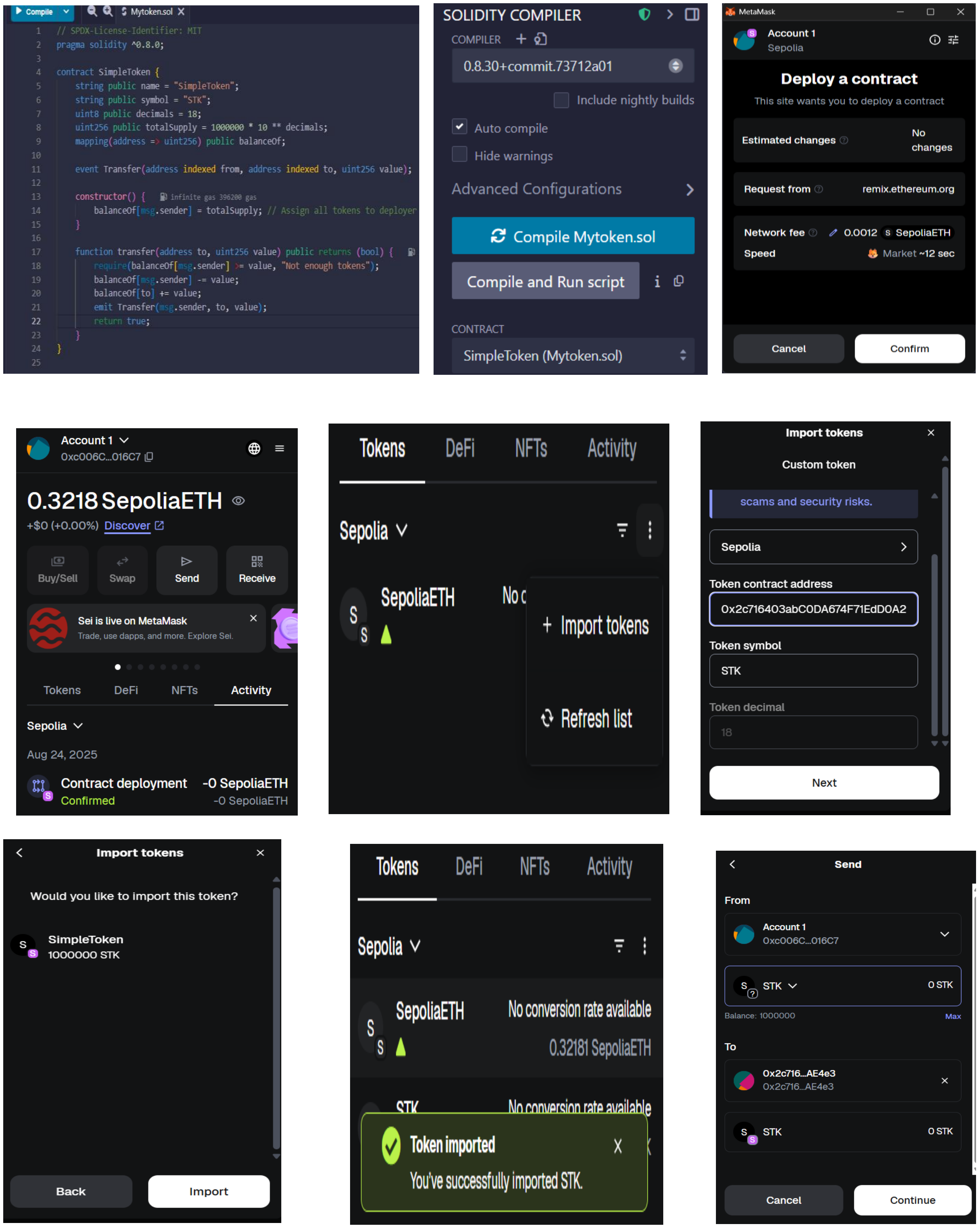
- Laptop
- Web Browser
- Remix IDE
- MetaMask Wallet
- Sepolia faucet

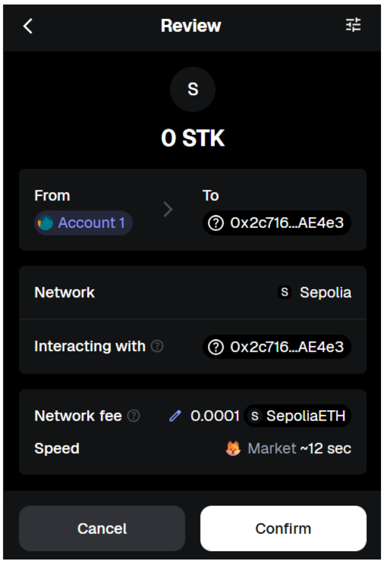
*** As applicable according to the experiment.**
Two sheets per experiment (10-20) to be used.

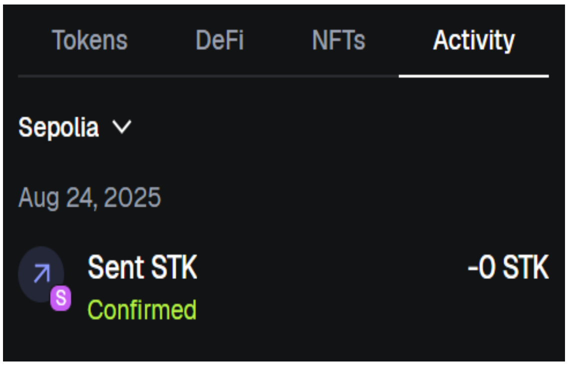
* Testing Phase: Compilation of Code (error detection)

NO ERROR

* Implementation Phase: Final Output (no error)







*Observation:

- **ERC-20** defines a standard for fungible tokens on the Ethereum blockchain, ensuring compatibility across wallets and dApps.

- It includes key functions like **transfer**, **approve**, and **transferFrom** for secure and standardized token operations.

- **Tokenization** enables representing real or digital assets as ERC-20 tokens for easy trading and ownership.

- Deploying ERC-20 tokens requires **Solidity smart contracts** and a **MetaMask wallet** for network interaction.

ASSESSMENT

Rubrics	Full Mark	Marks Obtained	Remarks
Concept	10		
Planning and Execution/Practical Simulation/ Programming	10		
Result and Interpretation	10		
Record of Applied and Action Learning	10		
Viva	10		
Total	**50**		

Signature of the Student :

Name :

Signature of the Faculty :

Regn. No. :

*** As applicable according to the experiment.***
Two sheets per experiment (10-20) to be used